

A man's profile is shown in a light, faded background. Overlaid on his head is a circular pattern of code snippets like 'uz.uz.uz' and 'uzn'. The main title is written in large, bold, black Japanese characters.

ウズウズカレツジ プログラマーコース

HAVING
はじまる!!

< 問題 >

テーブル『SAMPLE_4_4』から以下のデータを抽出するにはSQLをどう書いたらよい？

「学校別の最高得点・平均点を抽出（ただし、抽出対象は受験者数が3人以上の学校に所属している者とする）し、最高得点（降順）⇒平均点（降順）⇒学校名（昇順）の優先順で並び替える」

▼SAMPLE_4_4

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
2	CHOCO	M	MZMZ_COLLEGE	86
3	TARO	M	UZUZ_COLLEGE	86
4	RINRIN	F	« NULL »	93
5	POCHI	M	UKUK_COLLEGE	56
6	BEIBU	M	UZUZ_COLLEGE	52
7	POPO	F	UKUK_COLLEGE	86
8	BESU	M	MZMZ_COLLEGE	40
9	OMOCHI	F	UZUZ_COLLEGE	56
10	SASUKE	M	MZMZ_COLLEGE	20

< 答え >

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
WHERE SCHOOL_NAME IS NOT NULL  
      AND COUNT(*) >= 3  
GROUP BY SCHOOL_NAME  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC , SCHOOL_NAME
```

～テーブルの検索（SELECT）～

《 SELECT文の処理順 》

FROM ⇒ WHERE ⇒ GROUP BY ⇒ 関数 ⇒ SELECT ⇒ ORDER BY

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FROM句

どのテーブルの

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どの行を

GROUP BY句

どの集約キーでグループ化してデータ表を作り、

SELECT句

関数を動かし、どのカラムのデータを抽出して

ORDER BY句

どう並べて表示させるか

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関数はWHERE句の後に実行されるので
WHERE句の中では使えない！

< 答え >

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```

関数はWHERE句の後に実行されるので
WHERE句の中では使えない！

じゃあ集計結果に対する
絞りこみはできない？

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「学校別の最高得点・平均点を抽出（ただし、抽出対象は受験者数が3人以上の学校に所属している者とする）し、最高得点（降順）⇒平均点（降順）⇒学校名（昇順）の優先順で並び替える」

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< 答え >

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FROM SAMPLE_4_4  
  
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WHERE句の中では使えない！

じゃあ集計結果に対する
絞りこみはできない？

否！

HAVING

～テーブルの検索（SELECT）～

《 グループへの抽出条件（HAVING） 》

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
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```

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どの集約キーでグループ化して

データ表を作り、

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関数を動かし、どの行・どのカラムのデータを抽出して

SELECT句

どう並べて表示させるか

ORDER BY句

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GROUP BY SCHOOL_NAME  
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```

▼SAMPLE_4_4（抽出前）

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
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2	CHOCO	M	MZMZ_COLLEGE	86
3	TARO	M	UZUZ_COLLEGE	86
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5	POCHI	M	UKUK_COLLEGE	56
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「学校別の最高得点・平均点を抽出（ただし、**抽出対象は受験者数が3人以上の学校に所属している者とする**）し、最高得点（降順）⇒平均点（降順）⇒学校名（昇順）の優先順で並び替える」という意味のSELECT文になっている。

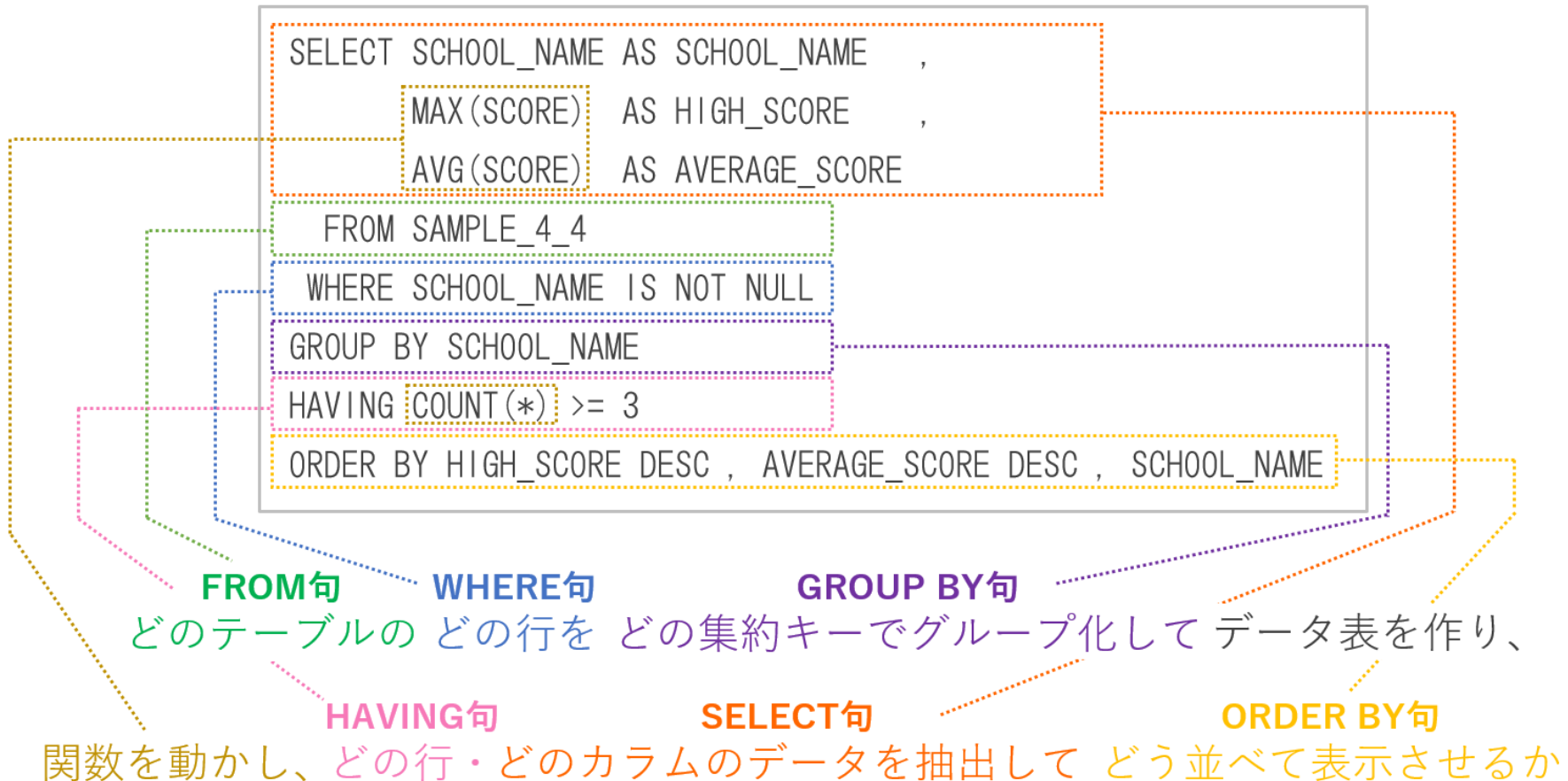
▼抽出結果

SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
UZUZ_COLLEGE	92	64.0000
MZMZ_COLLEGE	92	50.6667

～テーブルの検索（SELECT）～

≪ SELECT文の処理順 ≫

FROM ⇒ WHERE ⇒ GROUP BY ⇒ 関数 ⇒ HAVING ⇒ SELECT ⇒ ORDER BY



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STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE	MAX(SCORE)	AVG(SCORE)	COUNT(*)
1	MOCO	F	UZUZ_COLLEGE	56	92	64.0000	4
3	TARO	M		92			
6	BEIBU	M		52			
9	OMOCHI	F		56			
5	POCHI	M	UKUK_COLLEGE	56	90	73.0000	2
7	POPO	F		90			
2	CHOCO	M	MZMZ_COLLEGE	92	92	50.6667	3
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SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
MZMZ_COLLEGE	92	50.6667
UZUZ_COLLEGE	92	64.0000

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        , SCHOOL_NAME
```

SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
MZMZ_COLLEGE	92	50.6667
UZUZ_COLLEGE	92	64.0000



SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
UZUZ_COLLEGE	92	64.0000
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～テーブルの検索（SELECT）～

《 WHERE vs HAVING 》

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ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
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```

集約キーの絞り込みは
どちらに書いても
同じ結果になる！

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SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
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FROM SAMPLE_4_4  
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HAVING SCHOOL_NAME IS NOT NULL  
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```

SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
UZUZ_COLLEGE	92	64.0000
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SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
UZUZ_COLLEGE	92	64.0000
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～テーブルの検索（SELECT）～

≪ WHERE vs HAVING ≫

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
2	CHOCO	M	MZMZ_COLLEGE	86
3	TARO	M	UZUZ_COLLEGE	86
4	RINRIN	F	« NULL »	93
5	POCHI	M	UKUK_COLLEGE	56
6	BEIBU	M	UZUZ_COLLEGE	52
7	POPO	F	UKUK_COLLEGE	86
8	BESU	M	MZMZ_COLLEGE	40
9	OMOCHI	F	UZUZ_COLLEGE	56
10	SASUKE	M	MZMZ_COLLEGE	20

①

FROM句

WHERE句

GROUP BY句

どのテーブルの どの行を どの集約キーでグループ化して データ表を作り、

HAVING句

SELECT句

ORDER BY句

関数を動かし、どの行・ どのカラムのデータを抽出して どう並べて表示させるか

～テーブルの検索（SELECT）～

≪ WHERE vs HAVING ≫

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
  
FROM SAMPLE_4_4  
  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

WHERE句
なし

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
2	CHOCO	M	MZMZ_COLLEGE	86
3	TARO	M	UZUZ_COLLEGE	86
4	RINRIN	F	« NULL »	93
5	POCHI	M	UKUK_COLLEGE	56
6	BEIBU	M	UZUZ_COLLEGE	52
7	POPO	F	UKUK_COLLEGE	86
8	BESU	M	MZMZ_COLLEGE	40
9	OMOCHI	F	UZUZ_COLLEGE	56
10	SASUKE	M	MZMZ_COLLEGE	20

①

FROM句

どのテーブルの

②

WHERE句

どの行を

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≪ WHERE vs HAVING ≫

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       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
2	CHOCO	M	MZMZ_COLLEGE	86
3	TARO	M	UZUZ_COLLEGE	86
4	RINRIN	F	« NULL »	93
5	POCHI	M	UKUK_COLLEGE	56
6	BEIBU	M	UZUZ_COLLEGE	52
7	POPO	F	UKUK_COLLEGE	86
8	BESU	M	MZMZ_COLLEGE	40
9	OMOCHI	F	UZUZ_COLLEGE	56
10	SASUKE	M	MZMZ_COLLEGE	20



STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
3	TARO	M		92
6	BEIBU	M		52
9	OMOCHI	F		56
5	POCHI	M	UKUK_COLLEGE	56
7	POPO	F		90
2	CHOCO	M	MZMZ_COLLEGE	92
8	BESU	M		40
10	SASUKE	M		20
4	RINRIN	F	NULL	93

①

FROM句

どのテーブルの

②

WHERE句

どの行を

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≪ WHERE vs HAVING ≫

```
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       MAX(SCORE) AS HIGH_SCORE ,  
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       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE
1	MOCO	F	UZUZ_COLLEGE	56
3	TARO	M		92
6	BEIBU	M		52
9	OMOCHI	F		56
5	POCHI	M	UKUK_COLLEGE	56
7	POPO	F		90
2	CHOCO	M	MZMZ_COLLEGE	92
8	BESU	M		40
10	SASUKE	M		20
4	RINRIN	F	NULL	93



STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE	MAX(SCORE)	AVG(SCORE)	COUNT(*)
1	MOCO	F	UZUZ_COLLEGE	56	92	64.0000	4
3	TARO	M		92			
6	BEIBU	M		52			
9	OMOCHI	F		56			
5	POCHI	M	UKUK_COLLEGE	56	90	73.0000	2
7	POPO	F		90			
2	CHOCO	M	MZMZ_COLLEGE	92	92	50.6667	3
8	BESU	M		40			
10	SASUKE	M		20			
4	RINRIN	F	NULL	93	93	93.0000	1

①

FROM句

どのテーブルの

②

WHERE句

どの行を

③

GROUP BY句

どの集約キーでグループ化して データ表を作り、

④

HAVING句

関数を動かし、どの行・

SELECT句

どのカラムのデータを抽出して どう並べて表示させるか

ORDER BY句

～テーブルの検索（SELECT）～

《 WHERE vs HAVING 》

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE	MAX(SCORE)	AVG(SCORE)	COUNT(*)
1	MOCO	F	UZUZ_COLLEGE	56	92	64.0000	4
3	TARO	M		92			
6	BEIBU	M		52			
9	OMOCHI	F		56			
5	POCHI	M	UKUK_COLLEGE	56	90	73.0000	2
7	POPO	F		90			
2	CHOCO	M	MZMZ_COLLEGE	92	92	50.6667	3
8	BESU	M		40			
10	SASUKE	M		20			
4	RINRIN	F	NULL	93	93	93.0000	1



STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE	MAX(SCORE)	AVG(SCORE)	COUNT(*)
1	MOCO	F	UZUZ_COLLEGE	56	92	64.0000	4
3	TARO	M		92			
6	BEIBU	M		52			
9	OMOCHI	F		56			
2	CHOCO	M	MZMZ_COLLEGE	92	92	50.6667	3
8	BESU	M		40			
10	SASUKE	M		20			

①

FROM句

どのテーブルの

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WHERE句

どの行を

③

GROUP BY句

どの集約キーでグループ化して データ表を作り、

④

HAVING句

関数を動かし、どの行・

⑤

SELECT句

どのカラムのデータを抽出して どう並べて表示させるか

ORDER BY句

～テーブルの検索（SELECT）～

≪ WHERE vs HAVING ≫

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

STUDENT_ID	STUDENT_NAME	GENDER	SCHOOL_NAME	SCORE	MAX(SCORE)	AVG(SCORE)	COUNT(*)
1	MOCO	F	UZUZ_COLLEGE	56	92	64.0000	4
3	TARO	M		92			
6	BEIBU	M		52			
9	OMOCHI	F		56			
2	CHOCO	M	MZMZ_COLLEGE	92	92	50.6667	3
8	BESU	M		40			
10	SASUKE	M		20			



SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
MZMZ_COLLEGE	92	50.6667
UZUZ_COLLEGE	92	64.0000

①

FROM句

どのテーブルの

②

WHERE句

どの行を

③

GROUP BY句

どの集約キーでグループ化して データ表を作り、

④

HAVING句

関数を動かし、どの行・

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SELECT句

どのカラムのデータを抽出して どう並べて表示させるか

⑥

ORDER BY句

～テーブルの検索（SELECT）～

≪ WHERE vs HAVING ≫

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
MZMZ_COLLEGE	92	50.6667
UZUZ_COLLEGE	92	64.0000



SCHOOL_NAME	HIGH_SCORE	AVERAGE_SCORE
UZUZ_COLLEGE	92	64.0000
MZMZ_COLLEGE	92	50.6667

①

FROM句

どのテーブルの

②

WHERE句

どの行を

③

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どの集約キーでグループ化して データ表を作り、

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関数を動かし、どの行・

⑤

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⑥

⑦

ORDER BY句

～テーブルの検索（SELECT）～

《 WHERE vs HAVING 》

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
WHERE SCHOOL_NAME IS NOT NULL  
GROUP BY SCHOOL_NAME  
HAVING COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

```
SELECT SCHOOL_NAME AS SCHOOL_NAME ,  
       MAX(SCORE) AS HIGH_SCORE ,  
       AVG(SCORE) AS AVERAGE_SCORE  
FROM SAMPLE_4_4  
GROUP BY SCHOOL_NAME  
HAVING SCHOOL_NAME IS NOT NULL  
       AND COUNT(*) >= 3  
ORDER BY HIGH_SCORE DESC , AVERAGE_SCORE DESC  
       , SCHOOL_NAME
```

- ・ 結論、**集約キーへの絞り込みなどのWHERE句でできることはWHERE句です**意識を持ちましょう。
- ・ WHERE句はFROM句のテーブルのレコードを絞り込む場所、HAVINGはGROUP BYで集約したデータ表を絞り込む場所と、**役割の切り分け**を意識しましょう。
- ・ GROUP BYによる集約は重い処理（コンピュータへの負荷が高く処理速度が落ちる）であり、極力処理対象のデータを小さくして**処理速度を上げたい**、という意味でも早めに不要なデータを切り捨てることで処理速度を上げることのできるWHERE句での絞り込みを選択すべきでしょう。